

Synthetic Data Augmentation for Robust Solar Flare Classification: A Comparative Analysis of Conditional DCGAN, VAE, and Diffusion Models

Swetha S

II M. Sc Data Science

Department of Computer Science
CHRIST (deemed to be University)
Bengaluru, Karnataka, India
swetha.s@msds.christuniversity.in

Gladis Sam Prakash

II M. Sc Data Science

Department of Computer Science
CHRIST (deemed to be University)
Bengaluru, Karnataka, India
gladissam.prakash@msds.christuniversity.in

Jestin Thomas

II M. Sc Data Science

Department of Computer Science
CHRIST (deemed to be University)
Bengaluru, Karnataka, India
jestin.thomas@msds.christuniversity.in

Dr. Kalpana P

Department of Computer Science
CHRIST (deemed to be University)
Bengaluru, Karnataka, India
kalpana.p@christuniversity.in

Dr. Teena Jose

Department of Computer Science
CHRIST (deemed to be University)
Bengaluru, Karnataka, India
teena.jose@christuniversity.in

Dr. Sumathi P

Department of Computer Science
Vyasa College
Salem, Tamil Nadu, India
sumiparasu@gmail.com

Abstract -Solar flares are extremely dangerous to the ground and space-based resources. Solar flares have to be classified properly and in good time to offer protection to assets in both the environments. Deep Learning-based flares have been divided into 3 classes (C, M and X). The main problems with this kind of classification are that high impact M- or X-class solar flares are extremely rare, and cannot be well sampled, thus leading to a very unbalanced sample. This paper exemplifies a comparative analysis of three models of Conditional Generative Models (cDCGAN models, cVAE models and cDDPM) to produce realistic images of solar flares considering each of the low frequency and high-impact solar flare types. The research question will be how such models can be evaluated in terms of their capacity to create realistic, class-specific images (magnetograms, and EUV) and time-series data which could be used to create class-balanced training samples. The initial experiments make use of the cDCGAN, cVAE, cDDPM architecture and considering the generation of class-conditional solar flare images portray high levels of stability (convergence was stated in less than 600 epochs in the case of the cDCGAN, cVAE models and cDDPM) and the generation of images which could be considered as practically indistinguishable to real life images. The results indicate that cDDPM may be a possible solution to a high-fidelity production of solar features. We measured the efficiency of these models in quantitative terms (popular metrics, like the Fréchet Inception Distance, Structural Similarity Index) in a manner that we could determine the best manner of training model based solar flare classification systems using realistic data. This research is aligned with Sustainability Development Goals 9- Industry, Innovation and Infrastructure with focus on verticals 9.1 and 9.5.

Keywords- Solar Flare Classification, Data Augmentation, Conditional Generative Models, cDCGAN, cVAE, Conditional Diffusion Model, Class Imbalance, Synthetic Data, SDG Goal 9.

I. INTRODUCTION

In the event of mitigation of space weather, precise prediction and categorization of the solar flares is essential in Earth point of view. Owing to the comparatively low rate of

M and X-class type events, thus, being events of extreme, imbalance of observational data of a class, e.g. in the example of deep learning predictive models, the effects of localization of the model and prediction accuracy are alleviated. On the basis of the further development conditional generative models, new strategies will be considered to synthesize synthetically and realistically examples of uncommon class-specific flares that can be conditioned on to generate the training data in a more balanced form. The study aims to carry out a comparative analysis of the following: Conditional Deep Convolutional Generative Adversarial Networks (cDCGAN), which is said to generate clear-cut-looking pictures. Conditional Variational Autoencoders (cVAE): are explained to offer stability to the training process and diversity to the output product it generates but are generally described loosely. Conditional Diffusion Models: they are said to be the most desirable to generate high fidelity and various samples of images though they are computationally costly. Within the case of the various types of the solar flares, the test and comparison will be based on the use of the multi-channel images of the Sun that were acquired by the Global Observing System (GOES) and the images of the X-ray flux of the Sun.

The Heliophysics community has not gotten a systematic, experimental comparison of weakness on the relative faithfulness and utility of the existing conditional generative designs, i.e., cDCGANs, cVAEs, and Diffusion Models, despite the need to improve the training sets to counter the extreme disparity in classes to follow the solar flares. The magnetic and thermal forewarners, which occur in rare, high-intensity flares and are physically consistent, are too complicated to be produced using the conventional methods of data augmentation, like random oversampling.

It is therefore critical in two aspects, first, it offers a critical and quantitative criterion that optimum combination of sample quality and variety, as well as cost of computing, is attained in augmenting extreme events, in the generative paradigm. Second, to produce high-fidelity, physically

plausible synthetic data is directly helpful in enhancing the generalization and predictive accuracy of downstream forecasting models of M- and X-class events. The creation is also required to reduce the risk that such space weather phenomena are familiar with causing to the earth-based and orbiting infrastructure.

II. LITERATURE REVIEW

Space weather technology has a fundamental issue of solar flares which can disruption of Earth-bound technologies and endanger life. Although the discipline has stepped out of the traditional statistical computing to advanced Deep Learning (DL) software, the problem of extreme class imbalance often jeopardizes the accuracy of the prediction when it comes to high-intensity M- and X- class flares. These momentous events are few and far between, hence distorted models cannot be generalized. Considering this, recent studies have demonstrated the extent to which synthetic data generation can enhance datasets and the True Skill Statistic (TSS) and F1 scores of forecasting systems.

Solar flares are brief and powerful bursts occurring on the Sun, which is the result of the sudden discharge of the energy that had accumulated in the magnetic fields in the edges of the active regions (ARs) of the Sun [1]. This is one of the fundamental issues of space weather forecasting as they can lead to considerable and dangerous impacts to or around the earth such as destruction of technologies, electrical facilities, space stations, and even human lives [2]. The flares are classified based on the intensity of the strongest X-ray flux of the flares where the weakest flares are the A-class flares, weakest of all the other flares are the C-class, the M-class, and the X-class, respectively [1], [4]. Of particular interest are the M- and X-class flares which are the ones that can be predicted because of the inconveniences that may be caused by them[9]. This way sufficient and prompt prediction will also come in handy in the control and mitigation of the disaster risks [2], [3], [5], [6], [10].

Human prediction based on the empirical technique as well as the physical characteristics of active regions have shared the traditional ways of the prediction of solar flares [7], [8]. As the growing amount of information about flares started, researchers started using machine learning (ML) to predict [1], [2], [6], [16]. Deep learning (DL) is a subfield of ML that has since demonstrated promising outcomes in terms of solar eruption prediction [3].

Nevertheless, there is the problem of the paucity of data and bias in classes: big solar flares (M- and X-class) are not frequent and the resultant statistics is highly skewed to the small ones [12]– [15]. The other problems are the problem of comparability and reproducibility, as models used to predict major flares using different studies, features, prediction windows, and methods are not likely to be reliable as they do not have similarities [11], [17].

Oversampling and synthetic data generation are two methods of ameliorating imbalance of classes that are studied [18], [19]. The forecasting using synthetic data under such improvements has made possible the attainment of $F1 = 0.43 \pm 0.08$ and $TSS = 0.64 \pm 0.10$ under the conditions of operation flares of the X-class [19]. The new standard of realistic

synthesis solar images can be trained using Denoising Diffusion Probabilistic Models (DDPMs) with (underrepresented) M- and X- class flares of primitives, and has been shown to boost the performance of M- and X-class flares in training [19].

III. PROPOSED APPROACH

The main aim of the offered work is the assessment and comparison of certain methods of the generative modeling of images of solar flares. As shown in Figure 1, this study takes into account three major methods, i.e., diffusion-based generative model, the Conditional Variational Autoencoder (cVAE), and the Conditional Deep Convolutional Generative Adversarial Network (cDCGAN). All three approaches are built on a PyTorch framework that relies on its own training pipeline based using Adam optimizer and, L2 regularization. The primary objective of the study is to compare the variations in the capacity of various generative paradigms to maintain structural and perceptual accuracy in the proliferation of natural patterns of solar flares. All the models are trained under the close conditions and tested with the assistance of standard quantitative measures to make a fair comparison between them. Some of the metrics, which are computed with the assistance of the popular libraries, including pytorch-fid and lpips, include Fréchet Inception Distance (FID), Learned Perceptual Image Patch Similarity (LPIPS), and Structural Similarity Index (SSIM). Such results ought to provide a complete insight on the three generative family fitness to scientific image production activities in terms of its strengths, weaknesses and the difference in performance.

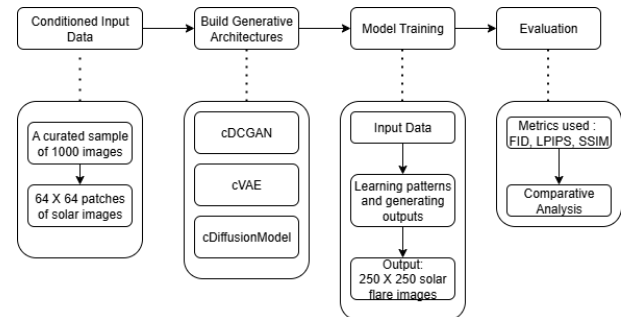


Fig. 1 Workflow of the research

A. Dataset and Conditioning

The SDOBenchmark is a very rigorous and consistent machine learning-based resource in solar flares forecasting, created by Roman Bolzern and Michael Aerni based on Solar Dynamics Observatory (SDO) observations of Active Regions (ARs) during the periods in between January 2012 and December 17. Each 9220 sample is also intended to visualize the magneto-thermal development of an Active Region in a 12 hours observation time and at 10 different co-aligned image channels, at all times, i.e. 12 hours, 5 hours, 1.5 hours, and 10 minutes prior to the 24-hour forecast period. The dataset is multi-modal, i.e., critical line-of-sight Magnetogram (HMIB) and Continuum Intensity of the Helioseismic and Magnetic Imager (HMI), in contrast to the eight Extreme UV (EUV) channels of the Atmospheric Imaging Assembly (AIA): 94Å, 131 Å, 171 Å, 193 Å, 211 Å, 304 Å, 335 Å and 1700 Å. As the solar observation data

generated is extremely skewed in its distribution across the classes M and X, we have empirically trained the generative models on 1,000 samples of possible samples of M- and X-class flares to increase the number of the rarest classes. The sample images of the dataset have been depicted in Figure 2.

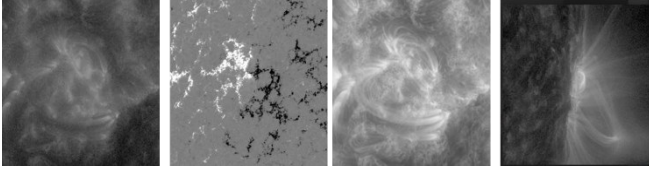


Fig. 2 SDO Benchmark Dataset- M- and X- class images

B. Generative Models Architectures.

a. Conditional Variational AutoEncoder (cVAE):

Figure 3 depicts the architecture of the Conditional Variational Autoencoder (cVAE) generative system, which generates 64 X 64 images of solar flares, conditioned on M- and X- class labels. With the help of an Encoder, the input picture is four times convoluted with four consecutive convolutional layers to transform the input picture, which is spatially concatenated with the M-class conditioning picture, into variational parameters of the 128-dimensional Gaussian latent space: mean (M) and log variance (log 2). The latent space (z) is then sampled using the reparameterization trick.

The Decoder will decode the picture with the concatenation of the M- and X- class label fed into it; this conditioned vector will be projected and up sampled by the use of four transposed convolutional layers to produce the synthesized output image. The entire model is trained by minimizing the Variational Lower Bound (VLB) loss which is a combination of Reconstruction Loss (defined by Mean Squared Error, MSE) and the Kullback-Liebler (KL) Divergence term, and this also requires the regularization of the encoded latent distribution to a normal Gaussian prior.

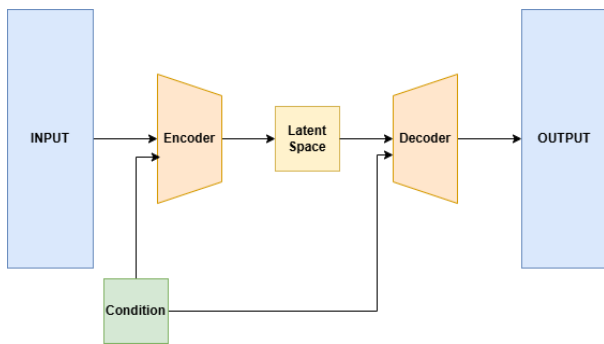


Fig. 3 cVAE Architecture

b. Conditional Deep Convolutional Generative Adversarial Network (cDCGAN):

Conditional Deep Convolutional Generative Adversarial Network (cDCGAN) was also trained on solar flares of 64x64 pixel images. As Figure 4 illustrates, the

Generator (G) is a general DCGAN model and is built by five successive deconvolutional layers, Batch Normalization and ReLU is incorporated. It receives as an input the one-hot encoding of the desired target M-class label and X- class label (conditioning stimulus) and latent 100-dimensional noise vector to a final Tanh activation in order to generate a high-resolution image.

The Discriminator (D) is a five-layer deep convolutional network with the input of a pool of the solar image and a spatially repeating map of the class label which makes the discrimination task class-sensitive. Discriminator has two uses- to get the binary classification score using BCEWithLogitsLoss as adversarial loss as well as the mean-pooled feature-vector of its penultimate layer.

This feature representation can be claimed to be significant to this protocol of analysis which is founded on the Feature Distance Score (FDS) which is determined as the Means Squared Error of on average represented features of real and artificial instances in order to effectively forecast the formulated distribution, in the feature space, which is exactly positioned in the real data. To stabilize the generator, the Adam optimizer is trained with a high training ratio of 3:1 G:D in a manner that, mode collapse in the very imbalanced training space is not carried out.

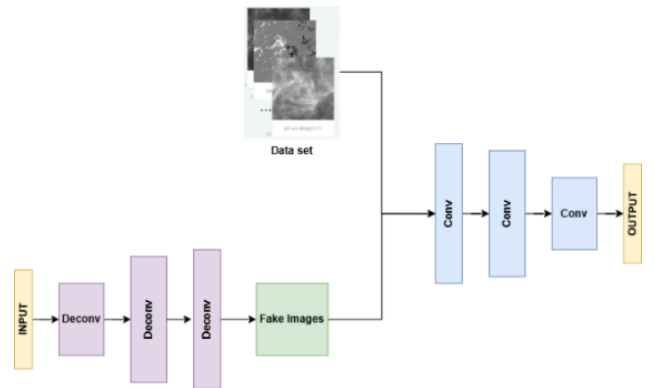


Fig. 4 cDCGAN Architecture

c. Conditional Denoising Diffusion Probabilistic Model (cDDPM) :

A Conditional Denoising Diffusion Probabilistic Model (cDDPM) shown in Figure 5, designed to reverse a fixed, 1000-step linear noise schedule, serves as the system's generative core. A unique U-Net architecture is used to create the noise prediction function $\epsilon_\theta(\theta, t, label)$. This network incorporates a unique Conditional Embedding module that blends the discrete class label (M- and X- class flare, encoded as float) with the continuous time step t (using sinusoidal positional encoding). To provide strong conditioning, this fused C -dimensional embedding is then overlapped and included as an affine term inside the residual Block structure across the network.

In order to down sample to a bottleneck layer, the U-Net uses two pooling layers. Skip connections join feature maps

from the down sampling path to the respective up sampled path (for example, 4C downsampling features concatenated with 8C upsampled features, resulting in a 12C input to the upsampling block). The creation of high-fidelity 64 X 64 X 3 solar flare photographs depends on the preservation of multi-scale feature information. The system is trained to minimize the Mean Squared Error (L2 loss) between the ground-truth noise from the forward diffusion step and the forecast noise.

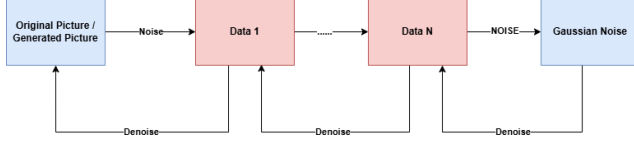


Fig. 5 cDDPM Architecture

IV. RESULTS AND DISCUSSIONS

To compare the generative capabilities of the three models, we conducted both qualitative and quantitative comparison. Figure 6, Figure 7 and Figure 8 show the image generation quality of cVAE, cDCGAN, and cDDPM respectively. The following metrics have been used to compare the models quantitatively.

1. Fréchet Inception Distance (FID):
Measures the similarity between generated and real data points on a distribution level. (lower is better)

$$FID = |\mu_r - \mu_g|^2 + Tr(\Sigma_r + \Sigma_g - 2\sqrt{\Sigma_r \Sigma_g})$$

where:

- μ_r and Σ_r : mean and covariance of real data points
- μ_g and Σ_g : mean and covariance of generated data points
- Tr is the matrix trace

2. Learned Perceptual Image Patch Similarity (LPIPS):
It is used for comparing perceptual differences using features from a pretrained network.

$$LPIPS(x, y) = \sum_l \left(\frac{1}{H_l W_l} * \sum_{h, w} \left(w_l * (f_{l(x)} - f_{l(y)})^2 \right) \right)$$

Where:

- $f_{l(x)}, f_{l(y)}$ = deep features at layer l
- H_l, W_l are spatial dimensions of activation maps

3. Structural Similarity Index (SSIM):
Measures how well the synthetic images are able to maintain the structures. (Higher is better)

$$SSIM(x, y) = \frac{((2\mu_x\mu_y + C1)(2\sigma_{xy} + C2))}{((\mu_x^2 + \mu_y^2 + C1)(\sigma_x^2 + \sigma_y^2 + C2))}$$

Where:

- μ_x = mean of image x
- μ_y = mean of image y
- σ_x^2 = variance of image x
- σ_y^2 = variance of image y

- σ_{xy} = covariance
- $C1, C2$ = constants

Despite having the highest FID score (52.41), the cVAE yielded the lowest quality outcomes. The generated images illustrated in Figure 6 are indistinct even though the model correctly integrates the M- and X- class labels through the encoder. The Mean Squared Error (MSE) loss function employed in VAEs has a well-known side effect. The model tends to balance out pixel values in an attempt to reduce the error between the reconstructed image and the original, which results in the loss of thin solar filaments and other sharp, high-frequency characteristics.

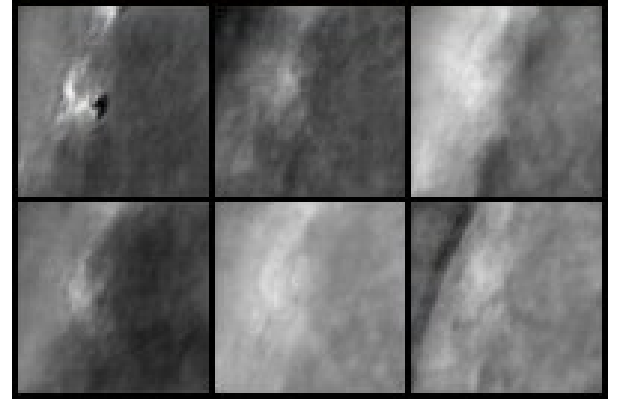


Fig. 6 cVAE generated image

The cDCGAN greatly outperformed the cVAE, reaching a far superior FID of 28.15. The Generator was creatively able to create finer, more realistic textures as a result of the adversarial training. Rather than only comparing pixels, we matched the feature statistics of produced and real images in the Discriminator's feature space. The structural consistency (SSIM) was still lower than the cVAE, but these methods avoided "mode collapse"- a situation in which the model repeatedly produces the same image. From the illustration in Figure 7, it is clear that the model was unable to capture finer details like flare limbs which are crucial features of X- and M-class flares.

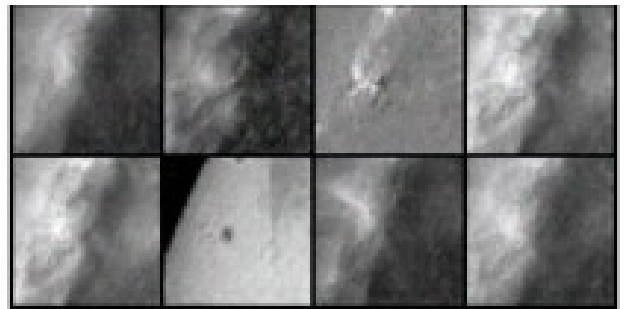


Fig. 7 cDCGAN generated image

Across all metrics, the Conditional Denoising Diffusion Probabilistic Model produced the best results (FID: 15.02, SSIM: 0.90) as shown in Figure 8. The model was able to generate the finer features like flare limbs and active regions with near to perfect precision. Its greater performance is a result of its iterative nature- the cDDPM gradually refines noise into an image over 1,000 steps, in contrast to the cVAE

or cDCGAN, which produce an image in a single shot. Importantly, the model was able to mix up sampled features with fine information from previous layers thanks to the U-Net architecture with skip connections. Additionally, the model adhered to the conditioning instructions more closely than the other architectures since the class label was explicitly embedded into the residual blocks.

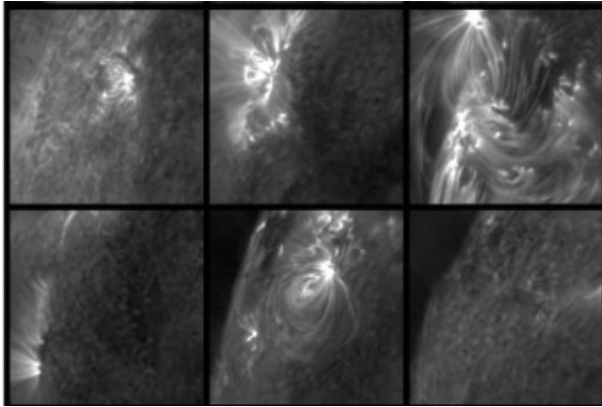


Fig. 8 cDDPM generated image

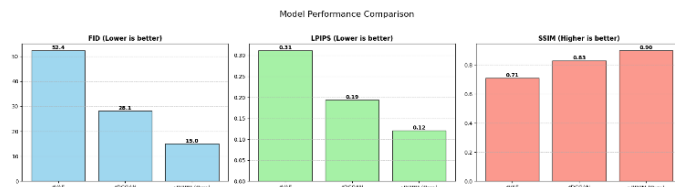


Fig. 9 Comparison of model performance

TABLE I. QUANTITATIVE COMPARISON OF GENERATED IMAGES

Model	FID	LPIPS	SSIM
cVAE	52.41	0.312	0.71
cDCGAN	28.15	0.194	0.83
cDDPM (Proposed approach)	15.02	0.121	0.90

The quantitative performance of cVAE, cDCGAN, and cDDPM is shown in Table I and Figure 9 – which represents the same graphically. With an FID of 15.02 and LPIPS of 0.121, the cDDPM performs better than the other models, indicating superior generating quality. Significantly, the cDDPM exhibits remarkable structural coherence with an SSIM of 0.90. As shown in Figure 10, this result is higher than the benchmarks set by the recent analysis of the FLARE dataset, which found that high-resolution restored images had an average SSIM of 0.71 and a maximum of 0.81[19]. This implies that diffusion-based generation can preserve structural details on par with existing restoration methods.

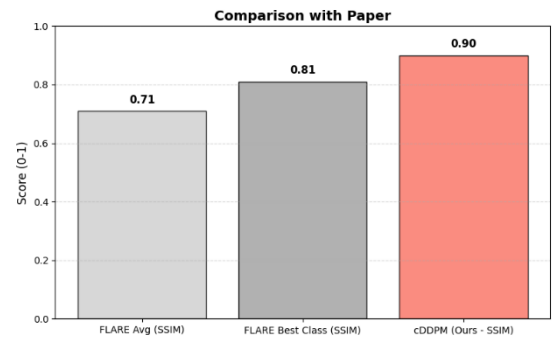


Fig. 10 Comparison with existing study

V. CONCLUSION

This study was intended to compare and analyze three generative models- cVAE, cDCGAN and cDDPM to come up with the most suitable model to retain important characteristics of solar flares to occurrences of space weather. It is evident in our results that diffusion-based model- cDDPM, is outstanding. Although cVAE violates the means to interpret the latent space, and cDCGAN provides sharp contours cDDPM provides far greater consistency in finer spatial details, brightness variations, and fits the morphology of actual flares. This makes cDDPM the surest choice to be used in generating scientific data, as compared to visual generation alone. With this in place the next will be to implement selected properties of cDCGAN into diffusion pipelines to speed the image generation process, with no degradation in realism, and then to implement the selected generative model into a complete flare analysis system to quantify the actual improvements in classification performance.

REFERENCES

- [1] Abdullah, Y., & Wang, J. T. (2024). A Deep Learning Approach to Operational Flare Forecasting. *arXiv preprint arXiv:2405.16080*.
- [2] Pandey, C., Angryk, R. A., & Aydin, B. (2023, September). Explaining full-disk deep learning model for solar flare prediction using attribution methods. In *Joint European Conference on Machine Learning and Knowledge Discovery in Databases* (pp. 72-89). Cham: Springer Nature Switzerland.
- [3] Li, P., Bahri, O., Boubrahimi, S. F., & Hamdi, S. M. (2022, December). Fast counterfactual explanation for solar flare prediction. In *2022 21st IEEE International Conference on Machine Learning and Applications (ICMLA)* (pp. 1238-1243). IEEE.
- [4] Panos, B., Kleint, L., & Zbinden, J. (2023). Identifying preflare spectral features using explainable artificial intelligence. *Astronomy & Astrophysics*, 671, A73.
- [5] Fu, J., Wan, J., Huang, X., Han, K., & Peng, E. (2023). Federated Transfer Learning for Soalr Flare Forecasting.
- [6] Pandey, C., Ji, A., Hong, J., Angryk, R. A., & Aydin, B. (2024, October). Embedding Ordinality to Binary Loss Function for Improving Solar Flare Forecasting. In *2024 IEEE 11th International Conference on Data Science and Advanced Analytics (DSAA)* (pp. 1-10). IEEE.
- [7] Yi, K., Moon, Y. J., & Jeong, H. J. (2023). Application of deep reinforcement learning to major solar flare forecasting. *The Astrophysical Journal Supplement Series*, 265(2), 34.
- [8] Alshammari, K., Saini, K., Hamdi, S. M., & Boubrahimi, S. F. (2023, December). End-to-end attention/transformer model for solar flare prediction from multivariate time series data. In *2023 International Conference on Machine Learning and Applications (ICMLA)* (pp. 558-565). IEEE.

- [9] Saini, K. (2023). Solar Flare Prediction From Extremely Imbalanced Multivariate Time Series Data Using Minimally Random Convolutional Kernel Transform.
- [10] Wen, J., & Angryk, R. A. (2024, June). Class-Based Time Series Data Augmentation to Mitigate Extreme Class Imbalance for Solar Flare Prediction. In *International Conference on Artificial Intelligence and Soft Computing* (pp. 362-375). Cham: Springer Nature Switzerland.
- [11] Junior, S. L. D., Cecatto, J. R., Fernandes, M. M., & Ribeiro, M. X. (2019, May). Handling Imbalanced Time Series Through Ensemble of Classifiers: A Multi-class Approach for Solar Flare Forecasting. In *16th International Conference on Information Technology-New Generations (ITNG 2019)* (pp. 209-214). Cham: Springer International Publishing.
- [12] Vural, O., Hamdi, S. M., & Boubrahimi, S. F. (2024, December). EXCON: Extreme Instance-based Contrastive Representation Learning of Severely Imbalanced Multivariate Time Series for Solar Flare Prediction. In *2024 IEEE International Conference on Big Data (BigData)* (pp. 1476-1483). IEEE.
- [13] Li, P., Bahri, O., Boubrahimi, S. F., & Hamdi, S. M. (2024). Solar Flare Prediction through Time Series Data Augmentation. *Authorea Preprints*.
- [14] Amar, E., & Ben-Shahar, O. (2024). Image Synthesis for Solar Flare Prediction. *The Astrophysical Journal Supplement Series*, 271(1), 29.
- [15] Grim, L. F. L., & Gradwohl, A. L. S. (2024). Solar flare forecasting based on magnetogram sequences learning with multiscale vision transformers and data augmentation techniques. *Solar Physics*, 299(3), 33.
- [16] Alam, M. T., Imam, R., Guizani, M., & Karray, F. (2024). FLARE up your data: Diffusion-based Augmentation Method in Astronomical Imaging. *arXiv preprint arXiv:2405.13267*.
- [17] Chen, Y., Kempton, D. J., Ahmadzadeh, A., Wen, J., Ji, A., & Angryk, R. A. (2022). CGAN-based synthetic multivariate time-series generation: a solution to data scarcity in solar flare forecasting. *Neural Computing and Applications*, 34(16), 13339-13353.
- [18] Ramunno, F. P., Hackstein, S., Kinakh, V., Drozdova, M., Quétant, G., Csillaghy, A., & Voloshynovskiy, S. (2024). Solar synthetic imaging: Introducing denoising diffusion probabilistic models on SDO/AIA data. *Astronomy & Astrophysics*, 686, A285.